GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

242491.1.pdf *Geochronology Record 1933*

242491.1: metamonzogranite, Piabubba Well

(Big Bell Suite, Austin Downs Supersuite; Youanmi Terrane, Yilgarn Craton)

SS Romano, MTD Wingate, Y LU, IOH Fielding and TJ Ivanic

Location and sampling

BYRO SG50-10, BOOLARDY 2244 Zone 50, MGA 482270E 7029601N
Warox Site TJIKAL190117 Sampled on 3 September 2019

This sample was collected from the northern edge of a 150 m-wide outcrop on Kalli Station, about 6.1 km south of Cardewandabubba Well, 4.2 km west-northwest of Wandarie Well, and 2.0 km north-northeast of Piabubba Well.

Geological context

The unit sampled is a metamonzogranite (Fig. 1) of the 2732–2686 Ma Big Bell Suite of the Austin Downs Supersuite (Van Kranendonk et al., 2013). The Big Bell Suite consists mainly of porphyritic and equigranular metagranodiorite to metamonzogranite, which commonly exhibits a magmatic foliation (Zibra, 2021: ENS report Big Bell Suite, viewed 3 August 2023). A metamonzogranite sampled about 21 km to the east yielded an igneous crystallization age of 2693 ± 7 Ma (GSWA 242490, Romano et al., 2023). Several granitic samples collected about 36 km to the north yielded igneous crystallization ages of 2717–2680 Ma (e.g. GSWA 105017, Nelson, 1996). A tonalite gneiss collected about 107 km to the southeast yielded an igneous crystallization age of 2724 ± 2 Ma (GSWA 185923, Wingate et al., 2011).

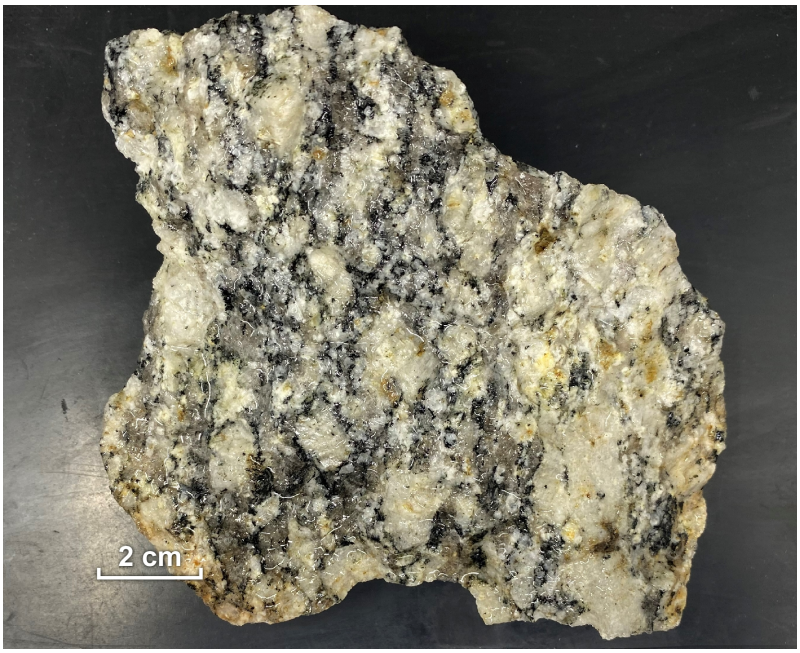

Figure 1. Hand specimen image (wet surface) for sample 242491: metamonzogranite, Piabubba Well

Petrographic description

The sample is an inequigranular, coarse-grained metamonzogranite, consisting of about 35–40% K-feldspar, 30% quartz, 25% plagioclase, 4% myrmekite, 3% biotite, <1% magnetite and accessory allanite and zircon. K-feldspar is subhedral, equant to tabular, microperthitic and up to 10 mm long. The larger grains contain tabular inclusions of plagioclase up to 1 mm in size, and many are fringed with lobes of myrmekite up to 0.3 mm in diameter. Quartz forms anhedral, irregular, recrystallized aggregates up to 7 mm long, in which individual grains are non-strained or weakly strained. Plagioclase is mainly anhedral and up to 7 mm in size. The rock contains patches of fine-grained, recrystallized plagioclase, mymekite, K-feldspar and quartz with polygonal texture. Green–brown biotite forms ragged, anhedral to subhedral laths up to 2 mm long, in part aligned in poorly defined, discontinuous folia. Magnetite forms small equant grains and irregular, elongate

grains up to 1 mm long. Allanite is anhedral, metamict, and up to 0.15 mm diameter. The sample exhibits a weak tectonic foliation. Deformation and grain-boundary recovery occurred at moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are pale violet to dark brown and mainly subhedral to euhedral. The crystals are up to 350 µm long, and equant to elongate, with aspect ratios up to 6:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, many crystals are dominated by high-U, metamict zones, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

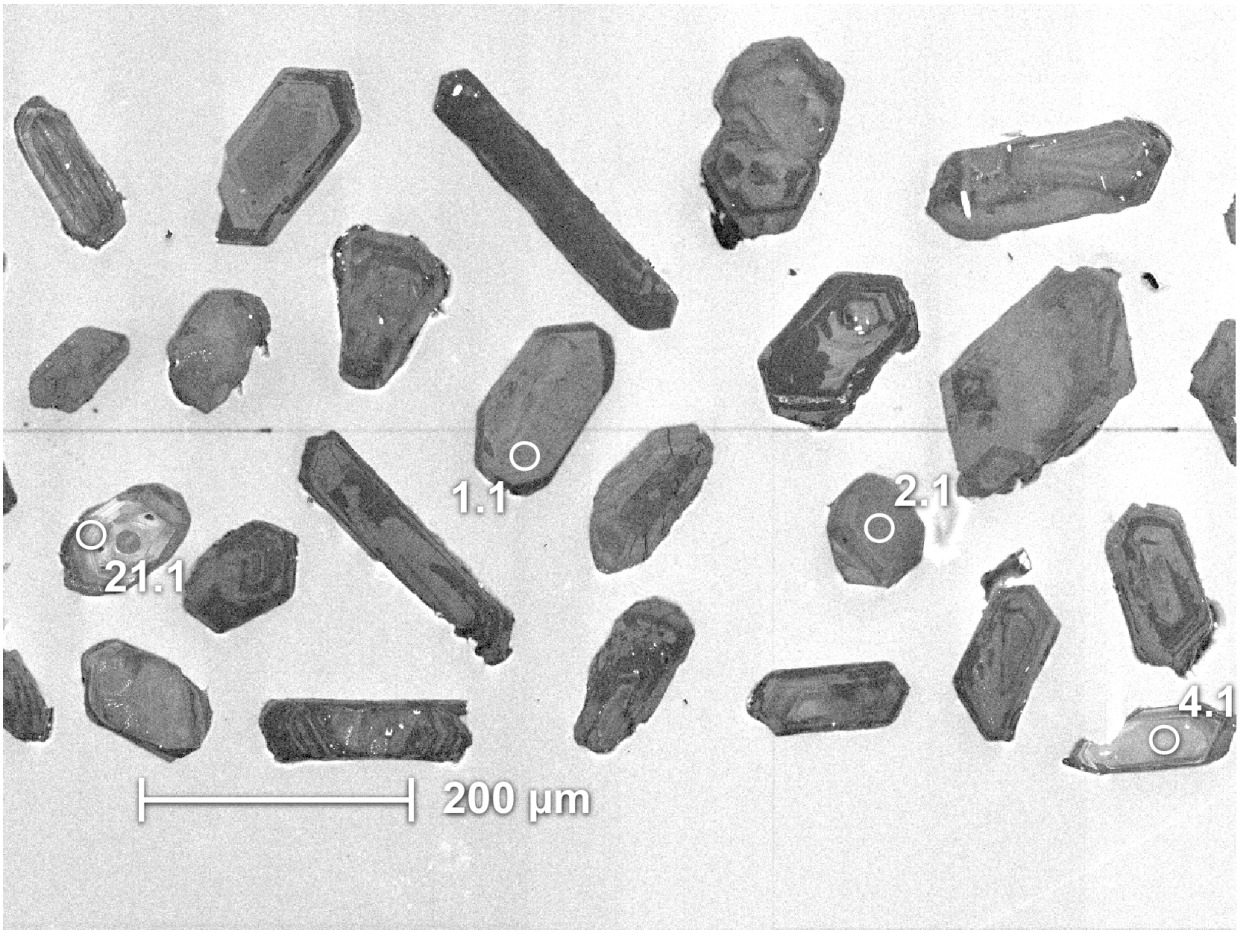

Figure 2. Cathodoluminescence image of representative zircons from sample 242491: metamonzogranite, Piabubba Well. Numbered circles indicate the approximate locations of analysis sites.

Analytical details

This sample was analysed on 28–29 October 2021, using SHRIMP-B. Eleven analyses of the M257 standard were obtained during the session, of which nine analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.70% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.22%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-one analyses were obtained from 21 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Two analyses are >5% discordant. The dates obtained from these two analyses (Group D; Table 1) are unreliable and are considered not to be geologically significant. The remaining 19 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their positions within the crystals.

Group I comprises 17 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2701 ± 5 Ma (MSWD = 1.4).

Group X comprises two analyses of zircon cores (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of c. 3296 and 3247 Ma.

The date of 2701 ± 5 Ma for the 17 analyses in Group I is interpreted as the magmatic crystallization age of the monzogranite. The dates of c. 3296 and 3247 Ma for the two analyses in Group X are interpreted as the ages of inherited zircons.

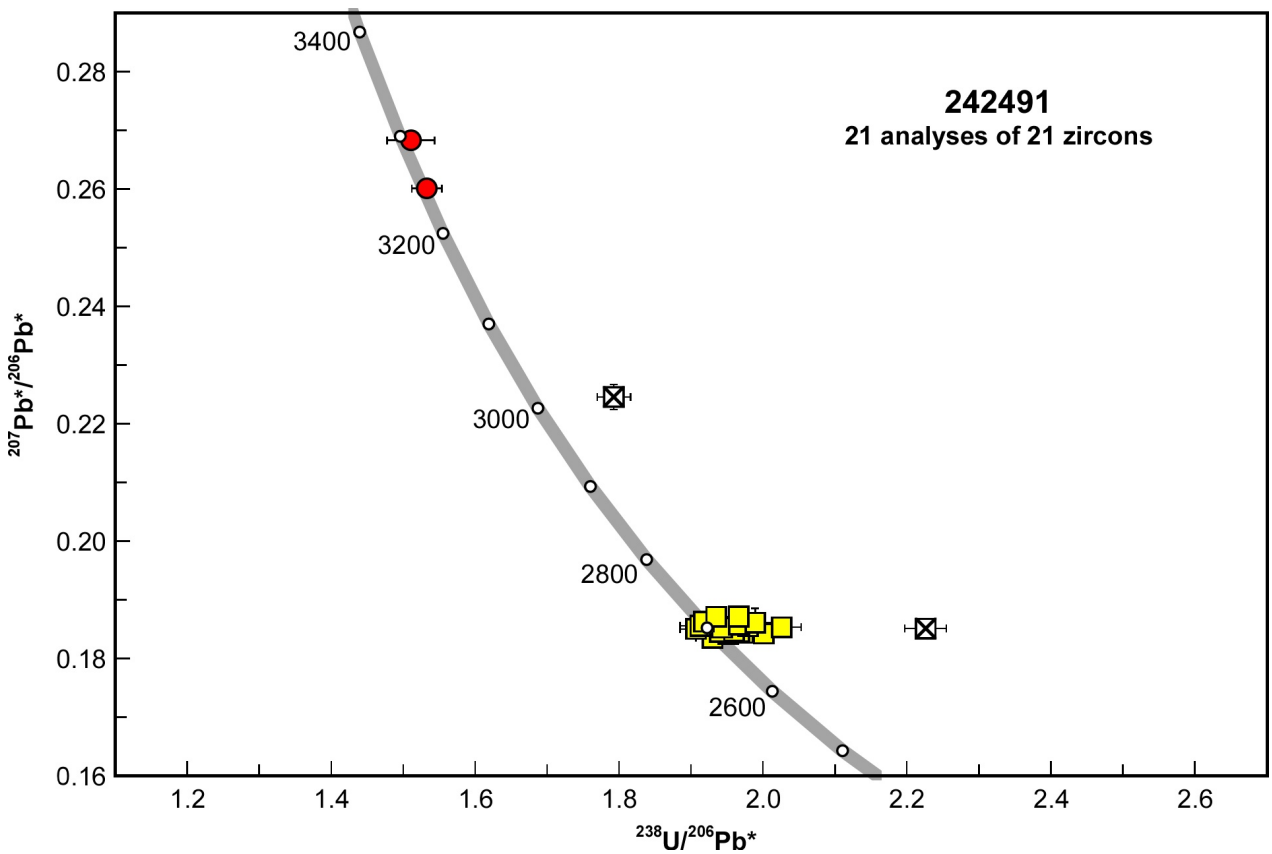

Figure 3. U–Pb analytical data for sample 242491: metamonzogranite, Piabubba Well. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircons); crossed squares indicate Group D (discordance >5%)

Links

Table 1: Ion microprobe analytical results for zircons from sample 242491: metamonzogranite, Piabubba Well: 242491.1.Table1.xls
SHRIMP spot images: 242491_spots.pdf
Introduction to geochronology information : Intro_2023.pdf

References

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Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Van Kranendonk, MJ, Ivanic, TJ, Wingate, MTD, Kirkland, CL and Wyche, S 2013, Long-lived, autochthonous development of the Archean Murchison Domain, and implications for Yilgarn Craton tectonics: Precambrian Research, v. 229, p. 49–92.

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Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

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Acknowledgment of Country

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